

OREF-INV-010: what the Boost cohort can and cannot answer about autosens

Scoping note

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Why this was scoped

INV-009 established that a sensitivity factor derived from a seven day total daily dose needs a real-time adjustment on top, and that per-person base error is large: entered settings run from 0.68 to 2.07 of what the 1800 rule gives, and 24% of people sit beyond what even a widened autosens clamp could absorb.

The design question that followed was whether autosens output can drive a slow loop that titrates the base. Testing it on Loop found nothing, but that test had two holes. Loop has no autosens, so nobody there is acting on one, and the reconstruction I used never reached the lower clamp, which left half the hypothesis untestable.

The Boost cohort has what Loop lacks: autosens ratios recorded by real devices. This note establishes what it can support before any work starts.

What is in the data

Two tables. `boost_decisions` holds 2.1 million per-decision rows from 36 users between August 2025 and August 2026. `boost_devicestatus_raw` holds the openaps block, including the recorded `sensitivityRatio`.

Quantity	Coverage
Recorded <code>sensitivityRatio</code>	200,118 readings, 11 users, 2026-07-18 to 2026-08-23
Profile sensitivity recorded	8 users, spans of 82 to 175 days
	3 (H with 12, tim with 9, B with 6)

Quantity	Coverage
Users with any profile sensitivity change above 10%	
Those users' time on dynamic ISF	99 to 100%
Users on profile sensitivity during the ratio window	2 (G at 0% dynamic, H at 27%)

Saturation is common and both bounds are reached, which the Loop reconstruction never managed:

User	Mean ratio	At ceiling	At floor
G	1.205	47.7%	0.0%
H	1.369	50.7%	6.4%
F	0.850	0.0%	20.2%
E	0.835	5.3%	18.7%
I	0.915	10.5%	16.7%
tim	0.971	0.0%	2.3%

What this cohort cannot answer

The question that prompted the scoping is not viable here. Asking whether autosens saturation precedes a change to the sensitivity factor requires people whose dosing sensitivity IS that setting, and who change it.

Those two groups do not overlap. Every user with a recorded profile sensitivity runs dynamic ISF for 99 to 100% of their decisions, so their profile value is a fallback rather than the number doing the work. The two users on profile sensitivity during the ratio window are G, who has no profile sensitivity recorded at all, and H, at 27%. That is one usable person.

Recruiting for it would need users on profile sensitivity with autosens, over enough months to catch settings changes. Nothing in the current cohort supplies that, and no amount of analysis will change it.

What this cohort can answer, in order of value

One: does the autosens reconstruction work at all

INV-009 built an autosens port and compared detectors with it. That port needed correcting three times: it used total delivered insulin where AAPS uses insulin net of the scheduled basal, it applied no meal exclusion to the four cohorts that record no carbohydrate, and it scored against an uncalibrated base. Each correction changed the conclusion, and the last one inverted it.

It has never been checked against a device. Here there are 200,118 recorded ratios from eleven people over five weeks, alongside the pump records the port consumes.

The work is to run the port over the same people and periods and compare, per person: correlation of the reconstructed ratio against the recorded one, agreement on which bound is being approached, and agreement on the fraction of time spent saturated. A port that tracks the device validates the INV-009 detector comparison. One that does not invalidates it, which is worth knowing either way and is cheap to establish.

This is the piece that decides whether the earlier work stands.

Two: how often does autosens actually saturate

Nobody appears to have characterised this. The table above suggests it is common and asymmetric, with two users pinned at a bound around half the time.

That matters directly for the clamp question INV-009 raised and could not settle. If a user spends half their time at the ceiling, the clamp is not a safety margin that occasionally binds, it is the operating point, and the loop is running a sensitivity the detector disagrees with for half the day. Whether that is protective or is suppressing a real signal is exactly what the outcome data alongside it can address, since `boost_cgm` carries paired glucose.

The work is descriptive: saturation rates per user, by time of day, by whether carbohydrate was recently entered, and paired against time in range and time below range during and after saturation episodes.

Three: what dynamic ISF and autosens do to each other

Nine of the eleven users run both at once. Dynamic ISF recomputes sensitivity from recent total daily dose several times an hour. Autosens then multiplies it from recent deviations. The two mechanisms are adjusting the same quantity from different evidence, and nothing establishes whether they agree, are independent, or oppose each other.

INV-009 found that the figure dynamic ISF uses swings by about 41% of a daily total inside a single record. If autosens is partly correcting that swing, the two are fighting, and the combination is doing something neither was designed for.

The work is to pair the recorded `sensitivityRatio` against the recorded `variable_sens` and `dynamic_isf`, per decision, and measure whether they move together or against each other.

This is exploratory rather than confirmatory. It has no preregistered prediction and should be reported as description.

Scope and cost

Phase one is the validation and is the gate. Roughly a day, reusing `inv009/sensitivity_detectors.py` against `boost_devicestatus_raw`. Phases two and three are each two to three days and should only start if phase one shows the reconstruction is sound, since both lean on the same machinery.

Data is local and no new extraction is needed. The 2026-06-26 refresh note in the extraction records says `extract_boost.py` is idempotent per user, so extending the `sensitivityRatio` window forward is a re-run rather than new work, and would widen phase one's overlap.

What would need saying in any output

The cohort is eleven people, self-selected, running a development branch, over five weeks. It supports statements about mechanism and about whether a reconstruction is faithful. It does not support population claims, and it is not the group to draw a clamp recommendation from on its own.